

Navya Nitin Maheshwari

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EDUCATION

Shiv Nadar University, Chennai

August 2024 - Present

B.Tech in Artificial Intelligence and Data Science | CGPA: 8.2 (Semesters I–IV)

EXPERIENCE

Flex – Chennai, Intern

May 2026 – July 2026

GraphRAG · Neo4j · GPT-5 · Groq (Llama-3.3-70B)

- **Project: HTS Code Mapping Using GraphRAG** - Built a **GraphRAG** pipeline over a **Neo4j** knowledge graph combining **GPT-5** and **Llama-3.3-70B via Groq** with semantic reranking to automate Harmonized Tariff Schedule (HTS) classification of engineering parts from shorthand descriptions.
- Presented the pipeline via a **Streamlit** dashboard, enabling analysts to review and correct classifications in real time.
- Built an evaluation harness across **868** real engineering shorthand samples; explored **QLoRA fine-tuning of Qwen 2.5-7B** for shorthand expansion via sequence-level distillation from GPT-5, and authored an internal technical paper documenting the pipeline and results.

CSIR–CEERI, Research Intern

Dec 2025 – Jan 2026

Python · TensorFlow · Keras · CNNs · NumPy · scikit-learn · Matplotlib

- **Project: Built and trained a LeNet-5-based CNN image classifier**, implementing an end-to-end pipeline covering preprocessing, training, evaluation, and hyperparameter tuning.
- Ran controlled experiments on model architecture and training strategy to study impact on classification performance.

PROJECTS

Paperwise — AI Reading Coach for Academic Papers

Jun 2026 – Present

React 19 · TypeScript · TanStack Start · Tailwind · Vercel AI SDK · Gemini 3 Flash · Zod · pdf.js

- Built an **LLM-powered** reading coach that guides students through research papers via highlight-to-explain, auto-generated prerequisite maps, and personalized reading plans — one engine powering three interaction modes.

Aviation Incident Report Pattern Finder

April 2026

Python · LangChain · Qdrant · HuggingFace · HDBSCAN · Streamlit · Ollama

- Built a RAG pipeline over **15,000+** NASA ASRS aviation safety reports using hybrid BM25 + dense vector retrieval with cross-encoder reranking.
- Indexed **15,445** embeddings (BAAI/bge-large-en-v1.5, 1024-dim) in **Qdrant** with metadata filtering by aircraft type, phase of flight, and date.
- Applied **HDBSCAN** clustering and KeyBERT topic extraction to surface latent safety themes across incidents; deployed a Streamlit dashboard with co-occurrence graphs and local LLM inference via Ollama (Llama 3.2 3B).

ShieldPay — Parametric Insurance for Gig Workers

Mar 2026 – April 2026

FastAPI · React · PostgreSQL · scikit-learn · XGBoost · Razorpay

- Built an AI-powered parametric insurance platform for Q-commerce gig workers that auto-compensates income loss from weather, AQI, and outage disruptions with zero manual claims.
- Shipped an **XGBoost** + time-series pipeline for income prediction, disruption probability scoring, and dynamic premium pricing across 3 tiers.
- Built a three-layer fraud detection system (GPS spoofing, claim velocity, weather cross-validation) with 0–100 confidence scoring; integrated **Razorpay** sandbox for real-time UPI payouts.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C

AI/ML: RAG, GraphRAG, Knowledge Graphs, LLMs, Vector Databases, Hybrid Retrieval (BM25 + Dense), Reranking, Embeddings, Fine-Tuning (LoRA/QLoRA), Model Distillation, Quantization, CNNs, Deep Learning, Model Evaluation

Frameworks & Libraries: PyTorch, TensorFlow, Keras, scikit-learn, LangChain, Hugging Face Transformers, FastAPI, React, Next.js, NumPy, Pandas, Matplotlib, Plotly

Databases, Tools & Platforms: Neo4j, PostgreSQL, Qdrant, SQLite, Firebase, Git, Docker, Groq, Ollama, Streamlit, REST APIs, Hugging Face Spaces, Linux

CERTIFICATIONS

- Deep Learning Specialization (Coursera, Andrew Ng)
- Programming in Java (NPTEL)
- Intro to RAG (LinkedIn Learning), Intro to SQL (LinkedIn Learning)
- Agentic AI, Ed Donner - *currently*

ACHIEVEMENTS & EXTRACURRICULAR

- 3rd Place, College Ideathon on Sustainability & Real-World Problem Solving — developed NGOConnect, an NGO–donor–volunteer coordination platform. Also: Badminton Team (regional & national-level) and Dance Team volunteer, Shiv Nadar University.